SCC.443 Forensic Report

1 Filesystem allocation size

The filesystem allocation size (bytes per cluster) is **2048 bytes** (2 KB). It was calculated using the values retrieved after parsing the BIOS Parameter Block (BPB) that is at the beginning of the image. It showed that the bytes per sector were 512 and there were 4 sectors per cluster.

2 Form of corruption in filesystem

From the analysis, it could be deduced that the root directory has been totally **wiped**, removing every entry and whenever that area of the image was accessed it returned entirely 0x00 bytes. The root directory starts at logical sector 68 in the image and its dimension is of 16384 bytes (32 bytes per entry, maximum of 512 entries). Luckily the two File Allocation Tables (FAT1 and FAT2) were intact and checked against each other to verify FAT1's integrity. The Data Area was also still readable. To summarize, it can be said that it was an **image quick format attempt**.

2.1 Number of files and directories

The total number of retrieved files is 57 (see RecoveredFiles folder). The number of retrieved directories is 8, along with 16 sub-directories. The files have been retrieved with the following methodology: after analysing the BIOS Parameter Block (BPB) and understanding the size of the clusters and sectors, following Microsoft Whitepaper on the FAT specifications[1], it was possible moving to the position of the root directory in the image. Once read, it was deduced that the root directory was empty, given that each of its bytes were 0x00. Since the root directory is empty, standard file extraction is impossible. A custom python script has been created to recover the data directly from the filesystem. First it checks that the FAT tables were not corrupted checking them against each other (FAT1 and FAT2), then it iterates through the FAT to rebuild the cluster chains (stopping when finding clusters that had EOF chars), next it reads these chains' raw data searching for standard directory entry headers. Following that, it continues with the parsing of the 32-byte directory entries to build a dictionary with the metadata of them (filename, starting cluster, size and a flag for sub-directories). Entries that were deleted (0xE5 deletion flag) or with long file names (0x0F) have been skipped (following the indications from the brief). Finally, the raw data of each recovered cluster chain, built from the starting cluster in the metadata dictionary, was extracted from the Data Area. The output for each file was truncated at the exact file size provided by the meta data to avoid padding/slack space in the final reconstructed files.

[1] https://academy.cba.mit.edu/classes/networking_communications/SD/FAT.pdf

3 Forensically relevant artifacts found

Out of the 57 recovered files, the most forensically relevant files found during the investigation were: *INVOICE.PDF* and *TRANSA-1.TXT*

- **INVOICE.pdf** contains an invoice in Russian language for the purchase of a DeWatt tool, generated by “Andrei Skottnikov". There is an address for a shopping centre in St. Petersburg, Russia. It lists then a crypto wallet address: *c0ffee254729296a45a3885639AC7E10F9d54979.*
- **TRANSA-1.TXT** on the other hand contains the hash of a cryptocurrency transaction happened between the user *Wes Clickwell* and *Andrei Skottnikov*. The receiver’s wallet address (Andrei Skottnikov) matches the one appearing in the invoice.

These files have been moved to a separate folder and hashed (see Evidence folder). All the other files were just basic Linux “man" files, for different tools. In the RecoveredFiles folder there are all the 55 remaining files and their hash value (checksum.md5)
To conclude, after working on the fat image its hash value (provided in the custody file, see Docs/custody-37078169-38.pdf) was checked and it matched.

```
sikri@vdi-scc-brw-016:~/h-drive/SCC.443/Programming - Assignment/Programs$
echo "566ea56db767c62c9dab8794d9b260e5 fat16-37078169-38.img" | md5sum -c -
fat16-37078169-38.img: OK
```